

Project Name	...
Online team meeting	https://tu-berlin.zoom-x.de/j/64327398600?pwd=iKF2e9D7PEM8Q552cgabakS0O6L9xa.1
Production system (if any)	http://amos.b-iq.net/
Test system (if any)	...
GitHub repository	https://github.com/amosproj/amos2025ss02-building-documentation-management-system
GitHub feature board	https://github.com/orgs/amosproj/projects/80/views/2
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/84
Team T-shirt (white)	https://www.shirtinator.de/loadBasket/vNV5XYUVMtg
Team T-shirt (black)	https://www.shirtinator.de/loadBasket/nQ9JYKa8F3W
Additional materials	...
Team mailing list	oss-amos-proj2@lists.fau.de

Last Name	First Name	GitHub User Name	Email Address
Kurtz	Daniel	daku-de	daniel.kurtz@fau.de
Deli	Deniz	ddeli	d.deli@campus.tu-berlin.de
Egge	Carl	carl-egge	c.egge@campus.tu-berlin.de
Halder	Arpita	arpita739	arpita.halder@fau.de
Kazi	Masudur Rahaman	kamsur	masudur.r.kazi@fau.de
Almalla	Nirmin	Nirmin96	nirmin.almalla@campus.tu-berlin.de
Ouabo	Samuel	so77nava	samuel.ouabo@fau.de
Billah	Masum	masum-abid	masum.billah@fau.de
Hussain	Syed Affan	syedaffan10	affan.hussain@fau.de
Djamen	Bastiane	bastydjamen	eugenie.djamen@fau.de

#	Meeting Day	Product Owner		Software Developer	Release Manager	Scrum Master	Comment
		Review	Planning				
1	2025-04-16		Daniel	Everyone else	N/A	Deniz	
2	2025-04-23	Daniel	Carl	Everyone else	Arpita	Deniz	
3	2025-04-30	Carl	Daniel	Everyone else	Masudur Rahaman	Deniz	
4	2025-05-07	Daniel	Carl	Everyone else	Samuel	Deniz	
5	2025-05-14	Carl	Daniel	Everyone else	Masum	Deniz	
6	2025-05-21	Daniel	Carl	Everyone else	Syed Affan	Deniz	
7	2025-05-28	Carl	Daniel	Everyone else	Bastiane	Deniz	Mid-term due
8	2025-06-04	Daniel	Carl	Everyone else	Arpita	Deniz	
9	2025-06-11	Carl	Daniel	Everyone else	Masudur Rahaman	Deniz	
10	2025-06-18	Daniel	Carl	Everyone else	Samuel	Deniz	
11	2025-06-25	Carl	Daniel	Everyone else	Masum	Deniz	
12	2025-07-02	Daniel	Carl	Everyone else	Syed Affan	Deniz	
13	2025-07-09	Carl	Daniel	Everyone else	Bastiane	Deniz	
14	2025-07-16	Daniel	Carl	Everyone else	Arpita	Deniz	Demo day!
15	2025-07-23	Carl	Daniel	Everyone else	Masudur Rahaman	Deniz	Retrospective
Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

Goals	Deliver a product that meets the stakeholder requirements.
	Everybody in the team wants to gain some new knowledge.
	Every team member should have fun during the project.
	Coming up with innovative solutions using modern technology.
	Develop our skills in teamwork and communication.
Meeting norms	We meet every Wednesday in the team meeting.
	POs will have weekly meetings with the stakeholder which are optional for the SDs.
	Everybody should be on time for the team meeting. If you are late or unavailable please inform the group beforehand through the defined communication channels.
	We practice a respectful and non-discriminative communication with all team members.
	Everyone should do their best to participate with Camera and Microphone turned on in the team meetings.
Working norms	Meetings will follow a clear structure that is quickly presented at the beginning of the meeting.
	We communicate or problems with the group and help colleagues that are stuck.
	We will follow a common Git Flow that will be defined by the SDs and follows industry best-practices.
	Commit often and use clear commit messages to avoid unnecessary conflicts.
	We will always use pull requests and code reviews instead of force pushes to improve coding quality.
Coordination norms	The POs are responsible for allocating assignments.
	The POs will alternately moderate the team meetings. The SM is responsible for keeping the team meetings on track.
Communication norms	We will use Discord as technical channel for programming topics.
	The Email list and Stand Up Email will be for organizational communication and announcements.
	Everybody should check these channels daily.
	We use english as a working language.
	The WhatsApp group is an additional optional channel for group conversations.
Consideration norms	No side conversation and technical conversations during the team meeting.
	For team decisions and disagreements an anonymous vote will be held and the majority decides.
Cont. improvement norms	Stand Up Emails and Happiness Index will be used to track team work.
	No pull requests should be accepted without feedback from at least one other team member (Positive feedback is welcome).
	We practice open feedback culture and continuously improve our working environment.
Rewards	By delivering on time everyone can reward themselves with a free weekend.
Sanctions	We define no sanctions.

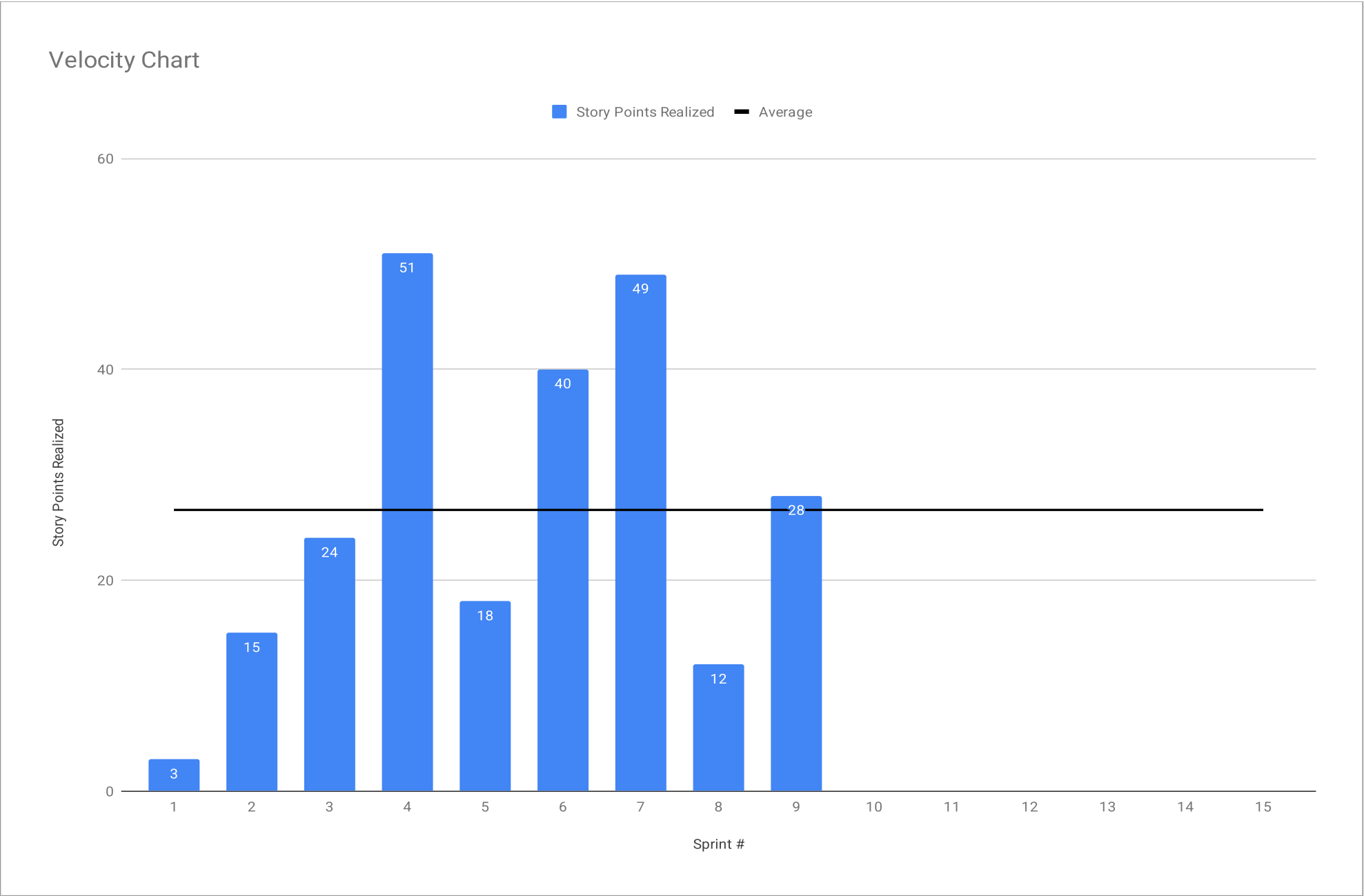
Signatures	
Scrum Master	Deniz Deli
Product owner	Daniel Kurtz
Product owner	Carl Egge
Software developer	Masudur Rahaman Kazi
Software developer	Bastiane Djamien
Software developer	Nirmin Almalla
Software developer	Arpita Halder
Software developer	Syed Affan Hussain
Software developer	Masum Billah
Software developer	Samuel Ouabo

Product Vision	Project Mission
Managing building portfolios often means navigating scattered, inconsistent document repositories. Our system enables facility managers to easily upload files, automatically extract and validate critical metadata, classify documents by type and building, and find exactly what they need through intuitive search and natural-language queries — reducing search times from hours to seconds. By integrating into broader building document management platforms, our solution enhances existing workflows and ensures seamless access to critical information across systems.	We aim to build a secure, multi-tenant backend that stores uploaded permits, certificates, and maintenance reports; applies AI-driven OCR and metadata extraction to classify and validate each document; and provides a web UI where users can query and filter their building records using plain language. By implementing and demonstrating a full workflow — from document upload to automatic processing and natural-language search on sample data — we will create the technical foundation for a production-ready system that significantly reduces the time and effort required to manage building documents.

Term	Definition
Document	A term to describe arbitrary files that contain information about a Building (e.g. Energieausweis, floor plan, ...)
Building	Corresponds to a real world building object and is described by one ore more Documents
Organization	An Organization is a high level data entity that is responsible for building one or more buildings and managing the corresponding data
User	A system user that belongs to one organization and interacts with the Document data
OpenAPI Client	Tool that handles interactions with APIs defined by OpenAPI specifications
File Explorer	Interface to browse Documents
Ollama	Tool for running LLMs locally or on premise
JWT	JSON Web Token used for secure user authentication
Chatbot	Interface to communicate with LLM
Energieausweis	Energy certificate showing a building's energy efficiency
Document Meta Data	Information about Document files including knowledge that has been extracted from the Documents content
Key Information	Each Document Category has a set of Key Information that are important and should always be automatically extracted on upload

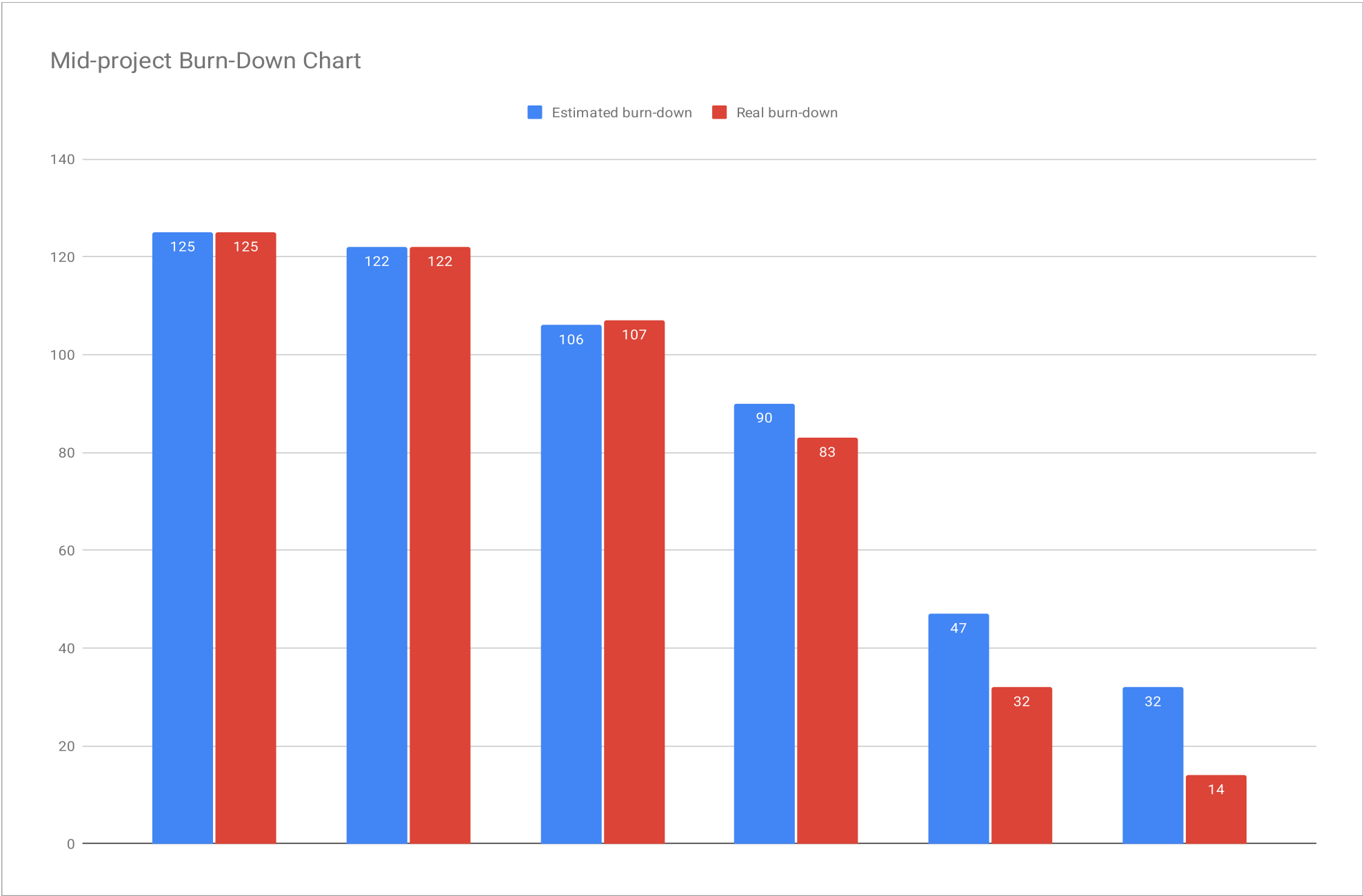
Sprint #	Sprint goal
1	Get to know the team and understand the stakeholder's requirements
2	Start scaffolding basic backend and frontend infrastructure and research domain-specific knowledge
3	Set up unified development and production environment through containerization and tooling
4	Fix the issue of sensitive documents in the repository and enhance backend with endpoints and database connection
5	Extend and prepare components so they can be easily connected
6	Prepare mid-project release by binding frontend and backend together and finalizing container orchestration
7	Refactor and enhance frontend to improve application data flow
8	Initiate automatic Building detection and general code refactorings
9	Classify Document automatically into a Building Document Category
10	Enhance frontend based on workshop and Collect list of relevant information per document category
11	Auto-Extract and display information from Document based on Document Category
12	AI chatbot can answer questions based on database information
13	Prepare Demo Day and Project Hand-Over
14	
15	

Sprint #	Story Points Realized	Average
1		326.66666667
2		1526.66666667
3		2426.66666667
4		5126.66666667
5		1826.66666667
6		4026.66666667
7		4926.66666667
8		1226.66666667
9		2826.66666667
10		26.66666667
11		26.66666667
12		26.66666667
13		26.66666667
14		26.66666667
15		26.66666667



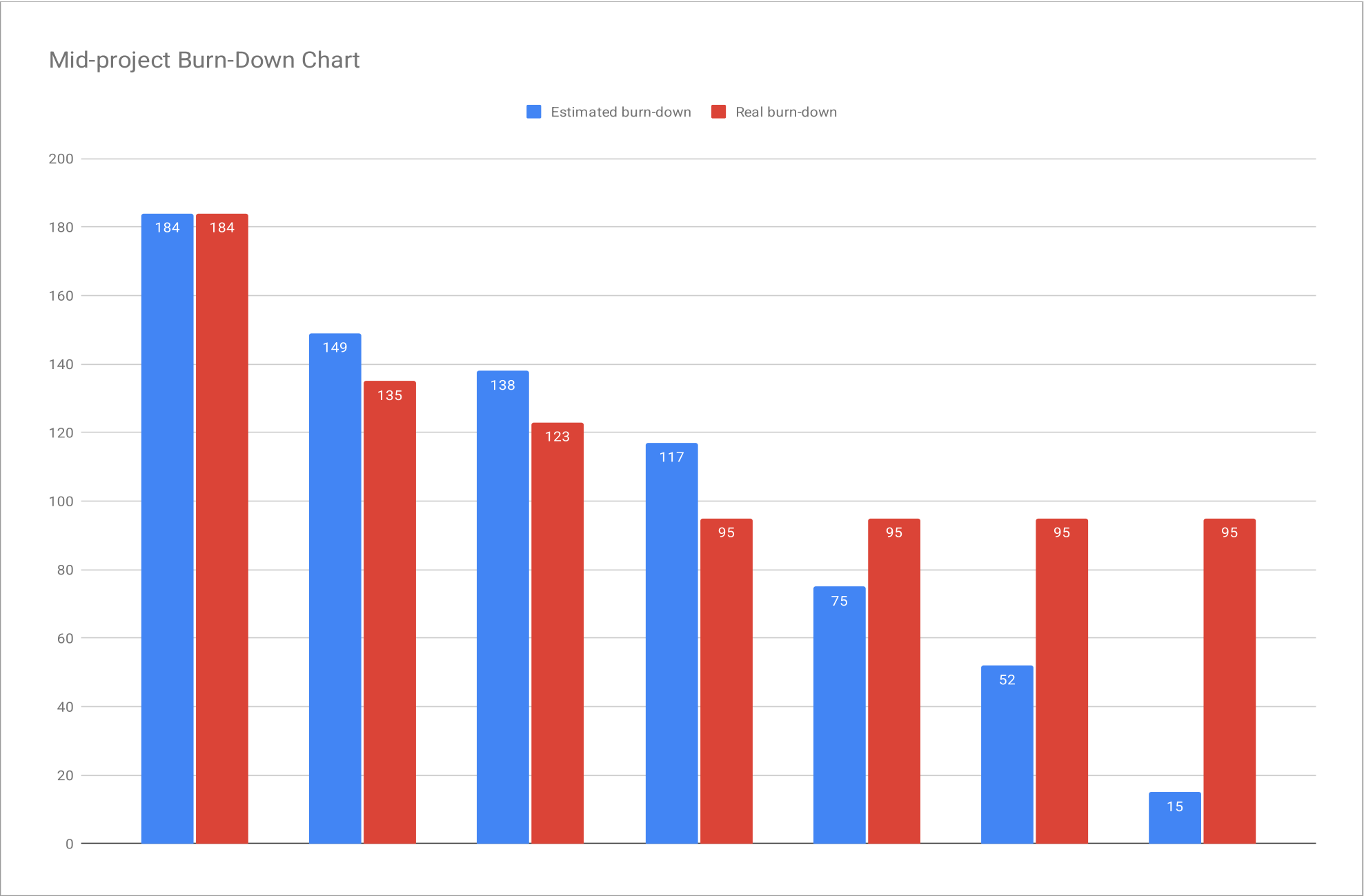
Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			125	125	151	151
Sprints						
1	Get to know the team and understand the stakeholder's requirements		3	125	3	125
2	Start scaffolding basic backend and frontend infrastructure and research domain-specific knowledge		16	122	15	122
3	Set up unified development and production environment through containerization and tooling		16	106	24	107
4	Fix the issue of sensitive documents in the repository and enhance backend with endpoints and database connection		43	90	51	83
5	Extend and prepare components so they can be easily connected		15	47	18	32
6	Prepare mid-project release by binding frontend and backend together and finalizing container orchestration		32	32	40	14
Features						
1	Get to know the team and understand the stakeholder's requirements					
		Align on Development Guidelines	1		1	
		Meet the industry partner	1		1	
		Create a team logo	1		1	
2	Start scaffolding basic backend and frontend infrastructure and research domain-specific knowledge					
		Create a shirt design	1		1	
		Initialize SW Bill of Materials	2		2	
		Create SW Architecture Description	3		3	
		Scaffold Basic C# Web API	3		2	
		Scaffold Basic Angular Application	3		3	
		Research Local RAG-Based LLM Solutions	2		2	
		Research typical Building Documents	2		2	
3	Set up unified development and production environment through containerization and tooling					
		Dockerize Front-End	2		3	
		Dockerize Back-End	2		2	
		Create DB Scheme Diagram	2		2	
		Create README	1		1	
		Consolidate Project Structure	2		2	
		Create Docker-Compose File	2		3	
		Linting and Formatting Frontend	1		1	
		Linting and Formatting Backend	1		2	
		Set Up GitHub Actions Pipeline	3		8	
4	Fix the issue of sensitive documents in the repository and enhance backend with endpoints and database connection					
		Fix: Remove Sensitive Building Documents	5		5	
		Implement Backend CRUD for Document	8		8	
		Implement Backend CRUD for Building	5		8	
		Integrate Linting into Pipeline	5		8	
		Add Authentication to Frontend	3		5	
		Design Frontend Mockup	5		8	
		Set Up Ollama	3		3	
		Implement Basic DB models in C#	8		5	
		Reorganize DB Scheme	1		1	
5	Extend and prepare components so they can be easily connected					
		Document Build Process in Video Format	2		5	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		Persist Building Data in DB	5		5	
		Query Ollama via Backend	5		5	
		Agree on DoD	3		3	
6	Prepare mid-project release by binding frontend and backend together and finalizing container orchestration					
		Upload and Store Document via Frontend	3		3	
		Document View Page displays Actual Documents	3		3	
		Set Up Apache Tika	5		5	
		Add Organization Entity to DB	8		8	
		Persist Data Using Docker Volumes	8		8	
		Implement Angular Frontend based on Mock Up	5		13	



Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			184	184	89	89
Sprints						
7	Refactor and enhance frontend to improve application data flow		35	184	49	184
8	Initiate automatic Building detection and general code refactorings		11	149	12	135
9	Classify Document automatically into a Building Document Category		21	138	28	123
10	Enhance frontend based on workshop and Collect list of relevant information per document category		42	117	0	95
11	Auto-Extract and display information from Document based on Document Category		23	75	0	95
12	AI chatbot can answer questions based on database information		37	52	0	95
13	Prepare Demo Day and Project Hand-Over		15	15	0	95
Features						
7	Refactor and enhance frontend to improve application data flow					
		Connect Frontend to LLM	3		3	
		Adopt Feature Branch Workflow and Clean Up Branches	2		2	
		Support OpenAPI Client Generation	8		13	
		Replace Default Web Services with Angular Frontend	3		5	
		Display JPG/PNG Files in Frontend	3		3	
		Implement Backend Logging	3		3	
		Refactor: Enforce OpenAPI SDK Usage	2		2	
		Seed Database with Organization and User	5		5	
		Apache Tika Integration during Upload	3		5	
		Bug in Production: CRUD Endpoints for Documents	3		8	
8	Initiate automatic Building detection and general code refactorings					
		Create a Building Form	5		5	
		Bug: Backend Container not working	2		3	
		Gather sample files of building documents	2		2	
		Bump tar-fs	2		2	
9	Classify Document automatically into a Building Document Category					
		Popup Window on Document Upload	5		13	
		Implement User Authentication in Backend	5		5	
		Initialize Document Category Metadata Extraction Schema	3		2	
		Refactor Building Entity	8		8	
10	Enhance frontend based on workshop and Collect list of relevant information per document category					
		Auto-Detect a Building from a Document	8			
		Enhance Frontend Design based on Team Design	5			
		Implement Session-Based Authentication for Frontend	3			
		GET Endpoint for Document Categories	3			
		Implement OCR using Tika	3			
		Auto-Detect Document Category from Uploaded Document	8			
		Enable Building and Category Change from Document View	5			
		Update Bill of Materials	1			
		Display Stored Documents in Explorer	3			
		Enforce HTTPS on Production Server	3			

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
11	Auto-Extract and display information from	Document based on Document Category				
		Adjust DB Scheme to Store Key Informations (depending on Category)	2			
		Display Key Information on File View Page	5			
		Auto-Extract Key Information on Document Category Change	5			
		Confirm multi-tenancy is respected in all UI/queries	5			
		Remove FastAPI from Ollama container	3			
		Remove Default weatherforecast Endpoint	2			
		Refactor Repository Structure and Cleanup	1			
12	AI chatbot can answer questions based on	database information				
		Enhance Documentation Wiki	3			
		Update README files	2			
		Semantic Search using Embeddings	8			
		Frontend: Add inline links to documents mentioned by Chatbot	5			
		Limit Chatbots Scope with Prompt Engineering	3			
		Enable Storing of Extracted Document Chunks	5			
		Implement Chunking of Document Contents	8			
		Remove unused tables from database (e.g. Document-Categories, -Permissions, -Tags)	3			
13	Prepare Demo Day and Project Hand-					
		Fine Tune Frontend	5			
		Fix Remaining Bugs	5			
		Prepare for Final Presentation	3			
		Prepare and reset Production Environment	2			



[illegible]

Type	Link / reference
User Documentation	https://github.com/amosproj/amos2025ss02-building-documentation-management-system/wiki/User-Documentation
Build and Deploy Documentation	https://github.com/amosproj/amos2025ss02-building-documentation-management-system/wiki/Build-&-Deploy
Technical Documentation	https://github.com/amosproj/amos2025ss02-building-documentation-management-system/wiki/Technical-Design

#	Context	Name	Version	License	Comment
1	UI	Angular	19	MIT License	Needed for developing UI as per client requirement
2	UI	Node.js	22.15.0	Custom (open-source)	Needed for Angular
3	UI	npm	10.9.2	Artistic License 2.0 with additions	Package manager, needed for Angular
4	Backend (Framework)	Microsoft.NETCore.App	Implicit	MIT License	.NET Core base runtime (System., Microsoft.)
5	Backend (Framework)	Microsoft.ASPNetCore.App	Implicit	MIT License	ASP.NET Core web framework (Controllers, Routing, Authorization)
6	Backend (Library)	Swashbuckle.AspNetCore	6.5.0	MIT License	Provides Swagger/OpenAPI documentation and UI
7	Backend (Library)	AspNetCore.HealthChecks.UI	6.0.4	Apache 2.0 License	Adds health check endpoints (/healthz) and optional health UI
8	Backend (Framework)	Entity Framework Core .NET Command-Line Tools	9.0.4	Apache 2.0 License	Used to manage EF Core database tasks from the terminal
9	UI	tar-fs	> 2.1.3	MIT License	Implicit dependency for frontend. Lower versions had security issues flagged by GitHub dependabot

Last Name	First Name	Value					
Halder	Arpita	1		1.00	OK		
Kazi	Masudur Rahaman	1					
Ouabo	Samuel	1					
Billah	Masum	1					
Hussain	Syed Affan	1		0	No size		
Djamen	Bastiane	1		1	Trivial size		
				2	Small size		
				3	Medium size		
				5	Large size		
				8	Very large size		
				13	Too large (size)		
How to play planning poker							
1. Everyone type their number into their value field, don't hit return yet							
2. Someone, perhaps a product owner, count down 3.. 2.. 1..							
3. Then, everyone hit return to submit their value							